

Machine Learning



Maximum a posteriori (MAP) Estimation

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https://github.com/safayani/machine_learning_course

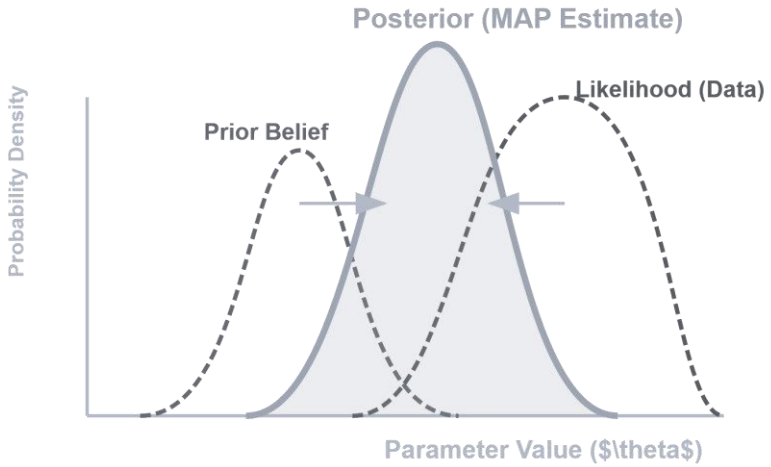
https://git.iut.ac.ir/safayani/courses/-/tree/main/Machine_Learning_course



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Introduction to Parameter Estimation

- How do we teach models to learn?
- The Goal: Given a dataset D, find the best parameters θ for our model.
- Standard Approach (MLE): Maximum Likelihood Estimation only looks at the data. It finds θ that makes the data most probable.
- The Problem with MLE: If D is small or noisy, MLE causes overfitting. It blindly trusts the data.
- The Solution: Incorporate existing knowledge (domain expertise) into the process.



The Foundation - Bayes' Theorem

- Injecting Prior Knowledge requires Bayes' Theorem:
- P(θ | D) is the Posterior: Belief about θ after seeing data.
- P(D | θ) is the Likelihood: How well θ explains the data.
- P(θ) is the Prior: Belief about θ before seeing data.
- P(D) is the Evidence: Probability of the data (normalization constant).

$$\underbrace{P(\theta|D)}_{a\ posterior\ probability} = \frac{\overbrace{P(D|\theta)}^{Likelihood} \quad \overbrace{P(\theta)}^{a\ prior\ probability}}{\underbrace{P(D)}_{Marginal\ probability}} \approx P(D|\theta)P(\theta)$$





Defining MAP Estimation

- Maximizing the Posterior:
- While MLE maximizes Likelihood $P(D \mid \theta)$, MAP maximizes the Posterior $P(\theta \mid D)$.
- Mathematically: $\theta_{MAP} = \text{argmax}_\theta P(\theta \mid D)$
- Substituting Bayes: $\theta_{MAP} = \text{argmax}_\theta [P(D \mid \theta) * P(\theta)] / P(D)$
- Since Evidence $P(D)$ does not depend on θ, we drop it:
- $\theta_{MAP} = \text{argmax}_\theta P(D \mid \theta) * P(\theta)$





The Log-Posterior Trick

- Making the Math Computationally Feasible:
- Multiplying many small probabilities causes numerical underflow.
- We apply the natural logarithm (log is monotonic).
- $\theta_MAP = \text{argmax}_\theta \log(P(D \mid \theta) * P(\theta))$
- Using $\log(a * b) = \log(a) + \log(b)$:
- $\theta_MAP = \text{argmax}_\theta [\log P(D \mid \theta) + \log P(\theta)]$
- Key Takeaway: MAP is the Log-Likelihood PLUS the Log-Prior!





MAP vs. MLE

- How do they compare?
- MLE: $\theta_{MLE} = \text{argmax}_\theta \log P(D \mid \theta)$
- MAP: $\theta_{MAP} = \text{argmax}_\theta [\log P(D \mid \theta) + \log P(\theta)]$
- Relationship:
- MAP is an extension of MLE.
- If we have NO prior knowledge (Uniform Prior), $\log P(\theta)$ is a constant.
- A constant doesn't affect the argmax, meaning MAP reduces exactly to MLE.





An Intuitive Example - The Coin Toss

- Why the Prior Matters:
- Scenario: Flip a coin 3 times, get 3 Heads. Estimate θ (probability of Heads).
- MLE Approach: θ_MLE = 3/3 = 1.0. Model assumes coin ALWAYS lands Heads (overfitting).
- MAP Approach: Introduce a Beta prior encoding belief 'Most coins are fair (θ ≈ 0.5)'.
- Result: MAP acts as a tug-of-war. Data pulls to 1.0, prior pulls to 0.5. Result: θ_MAP ≈ 0.7.
- As data increases (e.g., 10,000 flips), data overwhelms prior, MAP and MLE converge.





The Coin Toss

$$P(D|\theta)=\binom{N}{H}\theta^{H}(1-\theta)^{T}$$

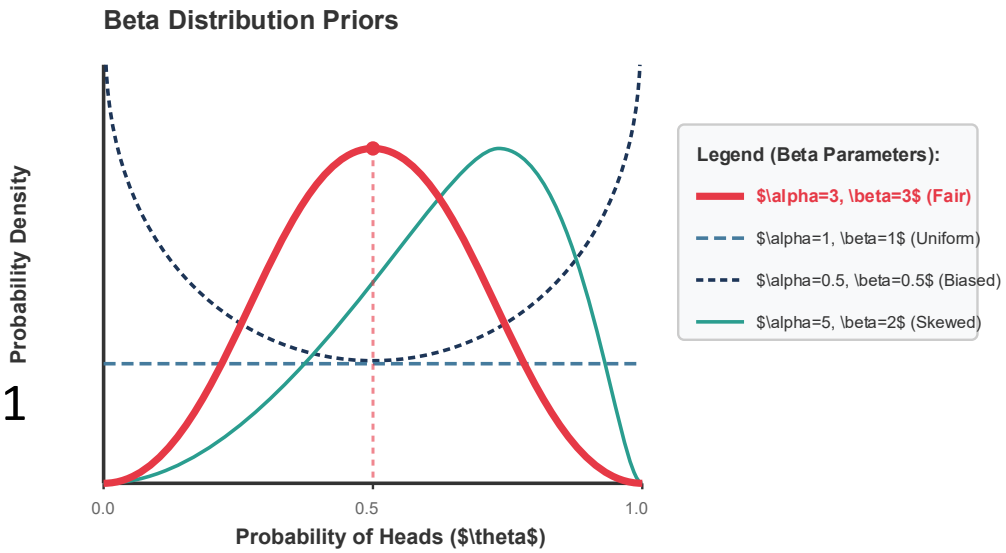
$$L=\log P(D|\theta)=H\log\theta \quad +T\log(1-\theta)$$

$$\frac{dL}{d\theta}=\frac{H}{\theta}-\frac{T}{(1-\theta)}=0$$

$\theta_{MLE}=\frac{H}{H+T}=\frac{H}{N}$ for H=3 and N=3 $\theta_{MLE}=1$

$p(\theta)=\theta^{\alpha-1}\theta^{\beta-1} assume\ \alpha=\beta=3$

$$P(\theta|D)=\theta^{H}(1-\theta)^{T}\theta^{\alpha-1}\theta^{\beta-1}$$



$\theta_{MAP}=\frac{H+\alpha-1}{H+T+\alpha+\beta-2}=\frac{5}{7}$ for H=3 , N=3 and $\alpha=\beta=3$

Let's assume the coin is actually slightly biased, and after 10,000 flips, we get 10,000 Heads ($H=10000, T=0$). Let's re-run the formulas:

$$\theta_{MLE}=\frac{H}{H+T}=\frac{10000}{10000}=1 \qquad \theta_{MAP}=\frac{10000+3-1}{10000+3+3-2}=\frac{10002}{10004}\approx 0.9998$$

MAP estimation Example

- We have observed data X = $\{x_1, x_2, \ldots, x_n\}$
  and we want to estimate the mean μ of a normal distribution. We ass
  ume a normal prior on μ with mean μ_0 and variance
  τ^2 , and we assume the data X is drawn from a normal distribution w
  ith mean μ and known variance σ2.

$$P(X|\mu) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$$

$$P(\mu) = \frac{1}{\sqrt{2\pi\tau^2}} \exp\left(-\frac{(\mu-\mu_0)^2}{2\tau^2}\right)$$

$$P(\mu|X) \propto P(X|\mu) \cdot P(\mu)$$

MAP estimation

$$P(\mu|X) \propto \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right)\cdot\exp\left(-\frac{(\mu-\mu_0)^2}{2\tau^2}\right)$$

$$\ln P(\mu|X) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2 - \frac{(\mu-\mu_0)^2}{2\tau^2} + \text{constant}$$

- Set the derivative to zero

$$\frac{d}{d\mu}\ln P(\mu \mid X) = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu) + \frac{\mu_0-\mu}{\tau^2} = 0$$





MAP estimation

$$\sum_{i=1}^{n} x_i - n\mu + \frac{\sigma^2}{\tau^2}(\mu_0 - \mu) = 0$$

$$n\mu + \frac{\sigma^2}{\tau^2}\mu = \sum_{i=1}^{n} x_i + \frac{\sigma^2}{\tau^2}\mu_0$$

$$\mu(n + \frac{\sigma^2}{\tau^2}) = \sum_{i=1}^{n} x_i + \frac{\sigma^2}{\tau^2}\mu_0$$

$$\mu = \frac{\sum_{i=1}^{n} x_i + \frac{\sigma^2}{\tau^2}\mu_0}{n + \frac{\sigma^2}{\tau^2}}$$





$$\mu_{\text{post}} = \frac{\sum_{i=1}^{n} x_i + \frac{\sigma^2}{\tau^2}\mu_0}{n + \frac{\sigma^2}{\tau^2}} = \frac{n}{n + \frac{\sigma^2}{\tau^2}}\,\bar{x} + \frac{\frac{\sigma^2}{\tau^2}}{n + \frac{\sigma^2}{\tau^2}}\,\mu_0$$

- **The posterior mean is a weighted average of the sample mean $\bar{x}$and the prior mean μ_0.**
- $\tau \to 0$**— highly concentrated prior:** The prior dominates; $\mu_{\text{post}} \to \mu_0$.
- $\tau \to \infty$**— highly diffuse prior:** The prior's influence vanishes; $\mu_{\text{post}} \to \bar{x}$.
- $\sigma^2 \to 0$**— negligible observation noise:** The data dominate; $\mu_{\text{post}} \to \bar{x}$.
- $\sigma^2 \to \infty$**— highly noisy observations:** The prior dominates; $\mu_{\text{post}} \to \mu_0$.

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$$\mu_{\text{post}} = \frac{\sum_{i=1}^{n} x_i + \frac{\sigma^2}{\tau^2}\mu_0}{n + \frac{\sigma^2}{\tau^2}} = \frac{n}{n + \frac{\sigma^2}{\tau^2}}\,\bar{x} + \frac{\frac{\sigma^2}{\tau^2}}{n + \frac{\sigma^2}{\tau^2}}\,\mu_0$$

- $n \to \infty$**, with** $\bar{x}$**and model parameters held fixed:** The data gain weight, and $\mu_{\text{post}} \to \bar{x}$.
- $n = 0$**— no observations:** The estimate is based on the prior: $\mu_{\text{post}} = \mu_0$.
- **Prior and data agree,** $\bar{x} = \mu_0$**:**$\mu_{\text{post}} = \bar{x} = \mu_0$, regardless of the weights.
- **Equal weights,** $n = \frac{\sigma^2}{\tau^2}$**:** $\mu_{\text{post}} = \frac{\bar{x}+\mu_0}{2}$.
- **Takeaway:** The data gain influence as the sample size increases or observation noise decreases. The prior gains influence as τ^2 decreases.





یک مدل احتمالاتی برای LS

فرض کنید داده های ما توسط مدل زیر تولید می شوند:

$$y_n = x_n^T \theta + \varepsilon_n$$

$\varepsilon_n \sim \mathrm{N}(\mu , \sigma^2)$:

ε_n یک نویز گاوسی با میانگین صفر و واریانس σ^2 است.
نویز با نمونه های تبدیل یافته جمع می شود و مستقل از نمونه هاست.

θ : پارامترهای مدل است

$$\mathrm{P}(y_n \mid x_n, \theta) = \mathrm{N}(x_n^T \theta , \sigma^2)$$

Ridge Regression as MAP estimator

فرض می کنیم که اجزای θ دارای توزیع گاوسین مستقل و یکنواخت با میانگین صفر و واریانس $\frac{1}{\lambda}$ است.

$$\boldsymbol{\theta}_{\text{ridge}} = \arg\max_{\boldsymbol{\theta}} \log p(\theta \mid \mathbf{X}, \mathbf{y})$$

$$\overset{(a)}{=} \arg\max_{\boldsymbol{\theta}} \log \frac{p(\mathbf{y}, \mathbf{X} \mid \theta)p(\theta)}{p(\mathbf{y}, \mathbf{X})}$$

$$\overset{(b)}{=} \arg\max_{\boldsymbol{\theta}} \log p(\mathbf{y}, \mathbf{X} \mid \theta)p(\theta)$$

$$p(\mathbf{y}, \mathbf{X} \mid \theta) = p(\mathbf{y} \mid \mathbf{X}, \theta)\, p(\mathbf{X} \mid \theta).$$

$$\overset{(c)}{=} \arg\max_{\boldsymbol{\theta}} \log p(\mathbf{y} \mid \mathbf{X}, \theta)p(\theta)$$

Ridge Regression as MAP estimator

$$= \arg\max_{\boldsymbol{\theta}} \log[p(\theta) \prod_{n=1}^{N} p(y_n \mid \mathbf{x}_n, \boldsymbol{\theta})]$$

$$= \arg\max_{\boldsymbol{\theta}} \log\left[\mathcal{N}\left(\theta \mid 0, \frac{1}{\lambda}\mathbf{I}\right) \prod_{n=1}^{N} \mathcal{N}(y_n \mid \mathbf{x}_n^\top \theta, \sigma^2)\right]$$

$$= \arg\max_{\boldsymbol{\theta}} \log\left[\frac{1}{\left(2\pi\frac{1}{\lambda}\right)^{D/2}} e^{-\frac{\lambda}{2}\|\theta\|^2} \prod_{n=1}^{N} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y_n - \mathbf{x}_n^\top \boldsymbol{\theta})^2}\right]$$

$$= \arg\max_{\boldsymbol{\theta}} -[\sum_{n=1}^{N} \frac{1}{2\sigma^2}(y_n - \mathbf{x}_n^\top \boldsymbol{\theta})^2 + \frac{\lambda}{2} \| \theta \|^2].$$

$$\arg\min_{\theta} \left[\frac{2\sigma^2}{N}\left(\sum_{n=1}^{N} \frac{1}{2\sigma^2}(y_n - \mathbf{x}_n^\top \boldsymbol{\theta})^2 + \frac{\lambda}{2}\|\boldsymbol{\theta}\|^2\right)\right]$$

$$\arg\min_{\theta} \left[\frac{1}{N}\sum_{n=1}^{N}(y_n - \mathbf{x}_n^\top \boldsymbol{\theta})^2 + \alpha\|\boldsymbol{\theta}\|^2\right]$$

$$\alpha = \frac{\lambda\sigma^2}{N}$$

Summary

- Key Takeaways:
- 1. MLE maximizes data likelihood but is prone to overfitting.
- 2. Bayes' Theorem combines data (Likelihood) with beliefs (Prior) for a Posterior.
- 3. MAP finds parameters maximizing this Posterior probability.
- 4. Formula: Log-Posterior $\propto$ Log-Likelihood + Log-Prior.
- 5. The Log-Prior acts precisely as a Regularizer (Gaussian = L2, Laplace = L1).



